Tanvi Ganesh Joshi

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Education

University of California, San Diego — San Diego, CA
MS in Computer Science and Engineering – GPA: 3.97/4.0 — *Sept 2024 – Jun 2026*

Indian Institute of Technology, Bombay — Mumbai, India
B.Tech. in Mechanical Engineering, Minor in CSE – CPI: 8.85/10.0 — *Nov 2020 – May 2024*

Experience

BeOne Medicines — June 2025 – Mar 2026
Advanced Analytics & ML Intern — *San Carlos, CA*

- Built production pipeline to score **10,000+ HCPs** from **1M+ biomedical interactions** ingested across **20+ data sources**.
- Redesigned KOL influence network incorporating Spoke-Hub referral dynamics, distinguishing local vs. regional/national KOL tiers; deployed unified entity model across **multiple oncology products**, scalable to full portfolio.
- Enriched social media data model via Meltwater API integration; automated Dataiku workflows and coverage audits, reducing manual analytics effort by **~70%** and preventing redundant third-party data purchases.

American Express — May 2023 – July 2023
Data Scientist, Decision Science & Strategy Intern — *Mumbai, India*

- Built and productionized end-to-end propensity model over **12M+ cardmember records** to predict B2B merchant transactions and select ad-targetable cohorts; engineered rule-based filters for high-precision training labels.
- Designed taxonomy mapping merchants, constructing ground-truth labels from raw transaction logs for supervised training.
- Implemented ML training pipeline with feature aggregation, normalization, PCA, and stratified splits; achieved **~0.80 ROC-AUC / ~0.70 F1**, producing deployable scored cohorts with **~2.5× lift**.

Projects

Multi-Agent LLM System with RL-Based Belief Inference (Avalon) | *Python, PyTorch* — 2026

- Built multi-agent LLM simulation with role-conditioned prompting, structured memory, and RL-based belief inference (suspicion + role prediction); supported by game engine with partial-information state tracking and APIs.
- Designed game loop enforcing partial-information constraints with discussion, voting, and team selection interfaces.
- Improved accusation accuracy to **0.73** and achieved **0.83** accuracy for low-information agents via RL policy learning.

Two-Tower Neural Recommendation System | *PyTorch, FAISS, Python* — 2026

- Built production-grade two-stage neural retrieval and ranking pipeline on MovieLens 1M (1M ratings, 6K users); dual-encoder trained with in-batch InfoNCE loss (2,047 negatives/positive) and L2-normalized embeddings indexed via **FAISS IVFFlat** (0.006ms/query, 5× speedup over flat index).
- Ranking MLP trained on hard negatives improved **NDCG@10 by 110%** and **MRR by 85%** over retrieval-only baseline; applied temporal per-user splits and evaluated with Recall@K, NDCG@K, and MRR.

Customer Churn Prediction with Causal Attribution | *XGBoost, SHAP, scikit-learn* — 2025

- Built XGBoost churn classifier on IBM Telco dataset (7K customers); engineered 5 domain features; achieved **AUC-ROC 0.849**, **68% precision** and **2.55× lift** at top-20% risk tier.
- Segmented churned population into 4 interpretable cohorts via K-Means in SHAP space; top 2 cohorts captured 75% of churners and **$257K of $327K** revenue at risk, enabling targeted retention campaigns projected to cut churn 18–22%.

Autonomous Driving Trajectory Prediction | *PyTorch, Python* — 2025

- Built Social-LSTM trajectory predictor for scenes with up to **50 interacting driving agents**; applied ego-centric normalization, rotation/mirroring augmentation, and physics-aware loss (MSE + direction + curvature).
- Reduced displacement MSE from **24.0→7.7** (**68% reduction**) via LR scheduling and early stopping.

Speaker Attribution using Sequence Models | *PyTorch* — 2024

- Built Transformer encoder-decoder with multi-head attention, masking, residual connections, and layer norm; trained speech segment classifier (**~85% accuracy**) and autoregressive LM (**~85 perplexity**) with causal masking.
- Tested AliBi positional encodings and analyzed attention maps to study long-range dependency learning.

Technical Skills

Languages: Python, SQL, HTML, LaTeX
Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, HuggingFace Transformers, sentence-transformers, LangChain
Data & Systems: Hive, Hadoop, Spark, Airflow, Dataiku, Ray, AWS (S3, EC2), FAISS, REST APIs
Focus Areas: NLP & LLMs, Ranking & Targeting, Predictive Modeling, Recommender Systems, Agentic AI, ML & DL
Miscellaneous: KVPY Fellow, DSC102 TA (UCSD), ME119 TA (IITB), Dept Academic Mentor (IITB), Grad Student Mentor (UCSD, IITB)